

LEUSHI, Russian Federation

WMO: 280640

Lat: 59.633N

Lon: 65.717E

Elev: 236

StdP: 14.57

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.9	-26.0	-36.3	0.7	-30.8	-31.2	1.0	-25.9	15.5	5.4	14.1	9.4	3.1	320	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	84.0	68.2	80.7	66.6	77.4	64.6	70.1	80.9	68.2	77.9	66.2	74.9	4.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	97.1	75.5	64.4	91.3	73.6	62.5	85.3	71.4	34.2	81.2	32.6	78.0	31.0	75.0	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	14.6	13.1	-35.6	89.4	7.9	2.5	-41.3	91.2	-46.0	92.7	-50.4	94.1	-56.2	95.9		
(5)			-35.8	73.0	7.9	2.6	-41.4	74.9	-46.0	76.4	-50.5	77.9	-56.2	79.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.2	0.4	5.4	22.5	37.2	49.5	60.6	64.8	59.5	49.2	35.2	18.0	5.9		
	DBStd	24.79	15.09	14.35	10.08	9.68	9.84	8.63	7.45	7.36	7.31	9.08	14.03	14.90		
	HDD50	7082	1538	1247	854	393	135	15	2	5	101	466	959	1367		
	HDD65	11437	2003	1667	1319	835	487	183	99	204	475	924	1409	1832		
	CDD50	1310	0	0	0	8	121	334	461	301	78	7	0	0		
	CDD65	191	0	0	0	0	8	52	94	35	2	0	0	0		
	CDH74	1507	0	0	0	0	104	409	739	244	11	0	0	0		
	CDH80	337	0	0	0	0	15	90	176	55	1	0	0	0		
	WSAvg	5.1	4.3	4.2	5.6	5.9	6.4	5.6	5.1	5.0	5.0	5.3	4.8	4.3		
	PrecAvg	19.2	0.9	0.7	0.7	1.0	1.5	2.4	3.0	2.7	2.4	1.5	1.3	0.9		
(15) Precipitation	PrecMax	27.2	1.8	1.4	1.4	2.8	3.2	5.3	7.0	7.0	5.4	3.6	2.9	1.7		
	PrecMin	12.4	0.3	0.1	0.1	0.2	0.3	0.2	0.4	0.2	0.5	0.6	0.2	0.1		
	PrecStd	3.7	0.4	0.4	0.4	0.6	0.9	1.2	1.7	1.3	1.3	0.7	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.6	36.4	48.6	66.2	82.2	87.0	87.4	85.9	74.7	62.5	44.0	35.6		
		MCWB	32.7	32.6	39.4	53.1	63.3	67.6	70.2	69.2	61.7	51.5	39.5	33.8		
	2%	DB	29.3	33.0	42.2	59.3	76.6	82.6	84.5	80.2	68.7	56.0	40.6	32.3		
		MCWB	28.1	30.7	34.9	48.9	59.5	66.9	69.2	67.2	58.9	48.1	37.8	30.8		
	5%	DB	24.6	29.1	38.8	55.2	71.9	78.9	81.6	75.9	64.1	51.2	36.9	29.0		
		MCWB	23.5	26.8	33.4	46.1	57.0	65.8	67.5	64.6	56.0	45.7	34.7	27.4		
	10%	DB	21.1	24.9	35.7	51.2	66.5	75.0	78.3	71.9	59.9	47.5	33.6	24.9		
		MCWB	19.9	22.5	31.5	43.0	54.0	63.1	65.7	62.8	53.8	43.7	31.8	23.4		
	0.4%	WB	32.9	33.3	39.8	54.9	64.1	71.4	72.6	71.2	65.3	52.3	40.9	34.0		
		MCDB	33.6	35.5	47.4	63.1	77.9	81.3	84.7	82.8	72.1	59.2	43.1	35.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.9	30.9	36.2	50.5	60.6	68.9	70.5	68.3	60.1	49.1	37.9	30.9		
		MCDB	29.0	33.2	41.2	57.3	75.6	79.4	81.8	77.7	66.8	54.8	40.7	32.3		
	5%	WB	23.6	27.1	33.7	46.8	57.9	66.6	68.9	66.0	56.9	46.2	34.8	27.7		
		MCDB	24.5	29.1	37.8	53.9	69.6	76.3	79.1	73.8	62.3	50.9	36.7	29.0		
	10%	WB	19.9	22.7	31.9	43.6	55.0	64.4	66.9	63.7	54.6	43.6	32.2	23.5		
		MCDB	21.1	24.6	35.3	50.3	65.1	72.8	76.0	70.5	59.1	47.4	33.5	24.8		
	5% DB	MDBR	11.8	13.9	14.9	16.0	18.0	16.4	16.1	15.0	12.9	10.2	9.9	11.2		
		MCDBR	12.3	12.6	16.1	21.5	25.5	22.0	21.4	20.6	19.0	16.0	9.9	10.2		
		MCWBR	11.9	11.2	11.3	13.8	13.7	10.8	9.5	10.0	10.9	9.9	8.6	9.8		
		MCWBR	12.5	12.1	14.4	20.2	23.3	20.1	20.1	19.3	17.1	14.9	9.9	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.266	0.304	0.360	0.396	0.394	0.400	0.413	0.420	0.365	0.334	0.288	0.257		
		taud	2.006	2.029	1.964	2.084	2.288	2.339	2.302	2.248	2.370	2.304	2.105	1.959		
	Ebn at Noon		153	212	234	252	264	264	257	243	239	206	145	107		
		Edh at Noon	16	27	40	42	37	36	37	36	27	21	14	11		
(44) All-Sky Solar Radiation	RadAvg		159	451	926	1345	1605	1734	1689	1242	748	376	151	83		
	RadStd		12	31	62	133	128	128	148	106	75	49	26	10		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page